

Alberto Reyes Castro

AI, Data Analysis & Technical Documentation



Portfolio

Portfolio Website

GitHub Projects

Kaggle Notebooks

LinkedIn Profile

Projects, reports, notebooks and technical documentation.

Specializations

• Machine Learning • Data Analysis • Python Programming • Scientific Computing • AIoT & TinyML • Technical Documentation • Scientific Writing • Automation • Model Evaluation • AI-Assisted Tooling

Skills

Python / pandas / NumPy / SciPy / Matplotlib / scikit-learn / XGBoost / Machine Learning / Data Analysis / Model Evaluation / Google Colab / Jupyter Notebooks / GitHub / TypeScript / Next.js / React / API Routes / Gemini API / Ollama / Markdown / LaTeX / Technical Documentation / Regex / Fuzzy Matching / AIoT / ESP32-S3

Languages

Spanish – Native
English – Advanced C1
Japanese – JLPT N3 level

SUMMARY

Physics graduate and MSc Artificial Intelligence Technology student with experience in Python, machine learning, data analysis, AIoT systems, TypeScript/Next.js prototyping, scientific computing and technical documentation. Strong background in analytical problem-solving, model evaluation, reproducible notebooks, AI-assisted tooling and structured technical reporting.

Currently pursuing a two-year, full-time MSc in Artificial Intelligence Technology with Advanced Practice at Northumbria University, London Campus. Achieved Distinction-level marks of 70+/100 across all completed first-semester final assessments and projects, including a machine learning fraud detection project awarded 85/100.

My profile combines technical understanding, scientific training and clear communication. I have developed projects involving imbalanced classification, model evaluation, AIoT prototyping, static notebook analysis, AI-assisted review tools, transcription automation, regex/fuzzy matching, LaTeX reports, Markdown-based documentation and reproducible notebook-based workflows.

I am seeking junior opportunities in AI, data analysis, Python automation, technical documentation, QA, research support and analytical roles, particularly in environments where rigorous evaluation, reproducible workflows, structured reporting and clear communication of technical findings are valued.

RELEVANT EXPERIENCE

Jul 2026–Present

Independent AI/ML Technical Project – ModelHawk

PERSONAL PORTFOLIO PROJECT · London, UK 📍

Built ModelHawk, a Next.js and TypeScript prototype for auditing machine learning notebooks. Developed static Jupyter notebook parsing, ML workflow signal detection, heuristic risk scoring, AI-assisted examiner notes using Gemini API with Ollama fallback, viva-style question generation and LaTeX-based PDF audit reports. The project is designed to support technical review, portfolio improvement and machine learning evaluation.



Mar 2025–May 2025

Student Internship – Department of Communication and Scientific Culture (UC3)

INSTITUTO DE ASTROFÍSICA DE CANARIAS (IAC) · La Laguna 📍

Contributed to bilingual science communication and outreach at the IAC. Developed a Python-based, AI-assisted transcription workflow with GPU-accelerated Whisper to generate segmented multilingual subtitles. Built a semi-automated text-matching pipeline using regex and fuzzy matching to link outreach articles with scientific publications. Supported outreach, audiovisual workflows, text processing and technical reporting in LaTeX.



EDUCATION

Jan 2026–Jan 2028

MSc Artificial Intelligence Technology with Advanced Practice

NORTHUMBRIA UNIVERSITY, LONDON CAMPUS · London, UK 🏛️

Distinction-level marks of 70+/100 across all completed first-semester final assessments and projects, including machine learning, AIoT, AI project development and technical reporting.



Sep 2018–Jul 2025

BSc in Physics

UNIVERSIDAD DE LA LAGUNA · Tenerife, Spain 🏛️

Academic Merit Convocation award. Final Degree Project completed in English with high distinction. Coursework included Python for scientific computing, LaTeX/Overleaf for academic reporting, basic LabVIEW in experimental techniques modules, and IRAF on Ubuntu in an astrophysical techniques elective. Final Degree Project included Rietveld and Le Bail refinement using FullProf.



Sep 2022–Oct 2022

Intensive Japanese Course (4 weeks)

EDUCATION FIRST LANGUAGE ABROAD · Tokyo, Japan 🏛️



Jul 2017

Intensive English Course (4 weeks)

EDUCATION FIRST LANGUAGE ABROAD · Brighton, UK 🏛️



SELECTED TECHNICAL PROJECTS

- 2026 ModelHawk – AI-Assisted ML Notebook Auditor** – Personal AI/ML technical prototype built with Next.js, TypeScript and API routes. Parses Jupyter notebooks, extracts code and markdown signals, detects machine-learning workflow patterns, generates heuristic scores for leakage risk, metric quality, reproducibility and portfolio readiness, and produces AI-assisted examiner notes using Gemini API with Ollama/local fallback support. Includes viva-style questions, suggested markdown improvements and professional LaTeX PDF audit reports.
- 2026 Credit Card Fraud Detection using Machine Learning** – Machine learning project focused on imbalanced classification, preprocessing, resampling strategies, model comparison, threshold selection and fraud detection evaluation using Python. Used pandas, scikit-learn, XGBoost, SMOTE and Google Colab. Distinction mark: 85/100. Portfolio includes report, GitHub repository and Kaggle notebook.
- 2026 AI-Enhanced Smart Home System** – Modular IoT prototype using ESP32-S3 room controllers, Wokwi, Blynk, Edge Impulse, Arduino-style programming, rule-based automation and lightweight environmental safety classification. Designed room-level logic and documented the system architecture, behaviour and evaluation.
- 2026 Computational Physics Numerical Methods** – Scientific Python project implementing interpolation, numerical integration, ODE solvers and stochastic simulation with applications to computational physics. Used NumPy, SciPy and Matplotlib.
- 2025 Adaptive Whisper Transcription Pipeline** – Portable CPU/GPU transcription workflow for audio and video files using faster-whisper, automatic hardware selection, safe input handling, audio normalisation, progress tracking and optional SRT subtitle generation.
- 2025 IAC Publication–Press Note Matching** – AI-assisted workflow for matching IAC scientific publications with press notes using text preprocessing, regex, fuzzy matching and reproducible notebook-based analysis.
- 2025 Final Degree Project – Ferroelectricity and Ferroelasticity** – Scientific project involving crystallographic analysis using neutron and X-ray diffraction data, Rietveld and Le Bail refinement using FullProf, amplitode analysis and long-form scientific reporting in English.

CERTIFICATES & ADDITIONAL TRAINING

- 2025** AI & Big Data in Biomedical and Healthcare Applications (300h) – DYM Canarias
- 2025** IELTS Academic UKVI (7.0/9.0, C1) – British Council
- 2025** NASA Open Science 101 – NASA
- 2024** Introduction to Data Science – Santander
- 2024** Artificial Intelligence and Productivity – Santander
- 2023** IDEAUP Open Innovation Training – Wrocław University of Science and Technology
- 2022** Cambridge Advanced (C1) – University of Cambridge
- 2022** Basic CPR & First Aid – Cabildo de Tenerife
- 2019** COEFIS Physics Student Congress – Universidad de La Laguna
- 2017** Scientific Week Certificate – CSIC

ACADEMIC PROJECTS & OUTREACH CONTRIBUTIONS

- 2025** *Final Degree Project: Structural Mechanisms of Ferroelectricity and Ferroelasticity in Compounds with the Formula: β' -RE₂(MoO₄)₃*. Rietveld and Le Bail refinement performed using FullProf. This project may be further developed into a scientific manuscript.
- 2025** *El eclipse parcial que recordó a Canarias el valor de mirar al cielo*, **Astronomía Magazine**
- 2025** *The WEAVE spectrograph captures the interaction of supernova SN 2023ixf with its surroundings one year after its explosion*
IAC News
- 2025** *Partial Solar Eclipse of March 2025*, **IAC News**

CONTACT DETAILS & LINKS

Alberto Reyes Castro

📍 London, UK

📞 UK Phone: (+44) 7311 991878

📞 WhatsApp: (+34) 683 472 456

✉ albertoreyescastro20@gmail.com

🌐 **Portfolio:** albertoreyescastro.github.io

[in](https://www.linkedin.com/in/alberto-reyes-ba5546238) [linkedin.com/in/alberto-reyes-ba5546238](https://www.linkedin.com/in/alberto-reyes-ba5546238)

🐙 github.com/albertoreyescastro

📊 kaggle.com/albertoreyescastro20